UNIT 5 · TOPIC 6(A)

Noise — Measurement, Effects and Exposure Limits

Syllabus topic: Effect of noise and vibration on work performance

Sound and its measurement (Liljedahl pp. 207–211)

- Sound = pressure vibrations. The human ear hears **16–20,000 Hz** (few adults perceive above ~11,000 Hz) (p. 207).
- Sound magnitude uses RMS pressure on a logarithmic scale (p. 209).
- Sound-level meters read **dB(A)** — A-weighting mimics the ear's frequency response. Nebraska tests have reported dB(A) at the operator's ear since 1971 (Unit 4, Topic 11).

$$\mathrm{SPL} = 20 \cdot \log(p/p_0)\ \mathrm{dB},\ p_0 = 0.00002\ \mathrm{N/m^2}\ (20\ \mu\mathrm{Pa}\ —\ \text{threshold of hearing})$$

Decibel arithmetic is logarithmic

- Two equal sources add **+3 dB**: 90 + 90 = **93 dB**.
- Removing an 81-dB source from a 90-dB total leaves **89.4 dB** (GenRad chart example, p. 211: difference 9 dB → correction 0.6 dB) — removing a small source barely lowers the total.

Tractor noise data (Nebraska, pp. 207–208)

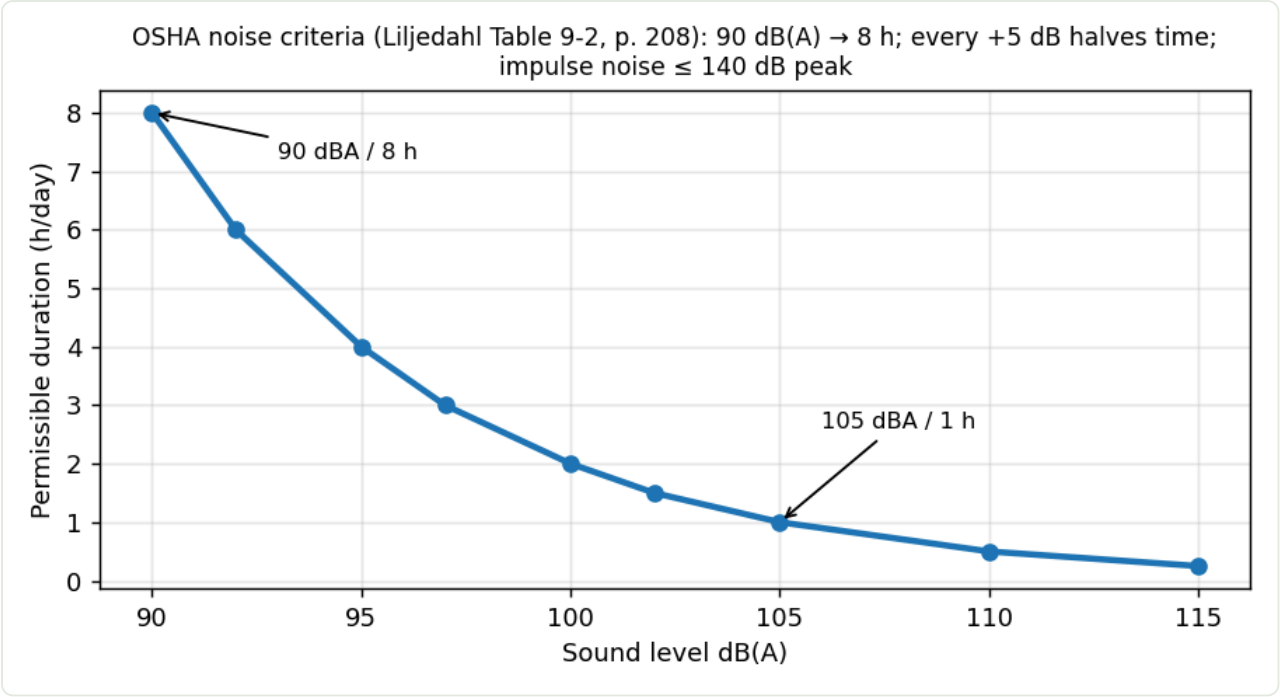
Condition (75% pull)	1974	1984
Without cab	95.2 dB(A)	94.0 dB(A)
With cab	91.4 dB(A)	79.6 dB(A)

Manufacturers achieved quiet operation mainly by engineering sound control into cabs.

Effects of noise on the worker

- Annoyance and rising fatigue.
- Masking of speech and warning signals — accident risk.
- Temporary threshold shift.
- With chronic exposure, PERMANENT noise-induced hearing loss (NIHL), starting near **4 kHz**.

Exposure limits (OSHA, Table 9-2, p. 208)



OSHA noise criteria — 90 dB(A) for 8 h; every +5 dB halves the time

Sound level dB(A)	Permissible time per day
90	8 h
95	4 h
100	2 h
105	1 h
110	½ h
115	¼ h

- Each +5 dB(A) **halves** the permissible time.
- Impulse noise ≤ **140 dB** peak.

Noise-control hierarchy

1. **Source:** muffler, engine covers, isolation mounts.
2. **Path:** cab, absorptive lining, distance.
3. **Receiver:** ear muffs/plugs (Topic 8).

Practice MCQs

Q36. The audible frequency range of the human ear is:

(a) 1–100 Hz
(b) 16–20,000 Hz
(c) 100–1000 Hz
(d) 20,000–50,000 Hz

Q37. The reference sound pressure for airborne sound (0 dB) is:

(a) 0.02 N/m²
(b) 0.00002 N/m²
(c) 2 N/m²
(d) 0.2 N/m²

Q38. Sound pressure level is defined as:

(a) $10 \cdot \log(p/p_0)$ (b) $20 \cdot \log(p/p_0)$

(c) p/p_0 (d) $p - p_0$

Q39. The 'A' in dB(A) denotes:

(a) Amplifier (b) Frequency weighting mimicking the human ear

(c) Average (d) Absolute

Q40. Two equal 90-dB sources together give:

(a) 90 dB (b) 93 dB

(c) 96 dB (d) 180 dB

Q41. Per OSHA (Liljedahl Table 9-2), 90 dB(A) exposure is allowed for:

(a) 2 h (b) 4 h

(c) 8 h (d) Unlimited

Q42. Each 5-dB(A) increase in level:

(a) Doubles allowed time (b) Halves the permissible exposure time

(c) No change (d) Quadruples it

Q43. Permissible daily exposure at 105 dB(A) is:

(a) 4 h (b) 2 h

(c) 1 h (d) 15 min

Q44. Impulse/impact noise must not exceed a peak of:

(a) 90 dB (b) 115 dB

(c) 140 dB (d) 200 dB

Q45. By 1984 Nebraska tests, tractors WITH cabs averaged operator noise of about:

(a) 95 dB(A) (b) 90 dB(A)

(c) 79.6 dB(A) (d) 70 dB(A)

Q46. Manufacturers reduced operator noise primarily by:

(a) Smaller engines (b) Sound-engineered cabs

(c) Slower speeds (d) Bigger mufflers only

Q47. Chronic noise exposure above limits causes:

(a) Temporary deafness only (b) Permanent noise-induced hearing loss

(c) Better hearing (d) No effect

Q48. An immediate work-performance effect of noise is:

(a) Better focus (b) Masking of speech and warning signals

(c) Cooling (d) Higher torque

Q49. The first priority in noise control is:

(a) Ear plugs (b) Reduction at the source

(c) Warning signs (d) Shorter breaks

Q70. Subtracting a small source (81 dB) from a 90-dB total leaves about (GenRad chart, p. 211):

(a) 9 dB (b) 81 dB

(c) 89.4 dB (d) 45 dB

Answers

Q36 — (b) 16–20,000 Hz — *Liljedahl p. 207 — ear range.*

Q37 — (b) 0.00002 N/m² — *Liljedahl p. 209 — p_0 = 0.00002 N/m².*

Q38 — (b) $20 \cdot \log(p/p_0)$ — *Liljedahl p. 209 — SPL = 20 log p/p_0.*

Q39 — (b) Frequency weighting mimicking the human ear

Q40 — (b) 93 dB
Q41 — (c) 8 h — *OSHA Table 9-2, p. 208 — the 90-dB(A)/8-h line.*
Q42 — (b) Halves the permissible exposure time — *Table 9-2 — the 5-dB halving rule.*
Q43 — (c) 1 h — *90 → 8 h, 95 → 4 h, 100 → 2 h, 105 → 1 h.*
Q44 — (c) 140 dB — *Impulse noise limit 140 dB peak.*
Q45 — (c) 79.6 dB(A) — *p. 207 — 1984 data: no cab 94.0 vs cab 79.6 dB(A).*
Q46 — (b) Sound-engineered cabs — *Quiet came from cab engineering.*
Q47 — (b) Permanent noise-induced hearing loss
Q48 — (b) Masking of speech and warning signals
Q49 — (b) Reduction at the source
Q70 — (c) 89.4 dB — *p. 211 — difference 9 dB → correction 0.6 → 90 - 0.6 = 89.4 dB.*

Sources: Liljedahl et al. — Tractors and their Power Units, Ch. 9 (noise pp. 207–211; vibration pp. 210–218; seating pp. 219–226); ISO 2631; OSHA criteria; CIAE Bhopal improved-tool studies; NPTEL ergonomics lectures.